MES' Institute of Management and Career Courses (IMCC)
(Autonomous) (A.Y. 2026-27)
Python Programming FYMCA Semester 1 (2026-28 batch)
Practical Assignment -1

Date – 22/10/2026

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student.txt

```
Shekhar
Sagar
Dipak
Priyanka
Roshni
Suraj
Akash
```

1. **Read a file containing student names (one per line) and store them in a list.**

```
with open("student.txt","r") as f:
    names = f.read().splitlines()

print(names)
```

Output:

```
['Shekhar', 'Sagar', 'Dipak', 'Priyanka', 'Roshni', 'Suraj']
```

2. **Print names longer than 5 characters**

```
with open("student.txt","r") as f:
    names=f.read().splitlines()

for name in names:
    if len(name) > 5:
        print(name)
```

Output:

```
Shekhar
Priyanka
Roshni
```

3. **Count how many names start with a vowel**

```
with open("student.txt") as f:
    names=f.read().splitlines()

count=sum(name[0].lower() in "aeiou" for name in names)
print(count)
```

Output:

```
1
```

4. **Save the filtered names (longer than 5 characters) into a new file.**

```
with open("student.txt") as f:
    names=f.read().splitlines()

with open("filtered.txt","w") as f:
    f.write("\n".join(name for name in names if len(name) > 5))
```

Output:

filtered.txt

```
1   Shekhar
2   Priyanka
3   Roshni
```

5. **Sort the list of names alphabetically and print them in ascending order.**

```
with open("student.txt") as f:
    names=f.read().splitlines()

for name in sorted(names):
    print(name)
```

Output:

```
Akash
Dipak
Priyanka
Roshni
Sagar
Shekhar
Suraj
```

6. **Print names that contain the letter "a".**

```
with open("student.txt") as f:
    names=f.read().splitlines()

for name in names:
    if "a" in name.lower():
        print(name)
```

Output:

```
Shekhar
Sagar
Dipak
Priyanka
Suraj
Akash
```

7. **Count word frequency in names (handle duplicates) and store results in a dictionary {name: count}.**

```
with open("student.txt") as f:
    names=f.read().splitlines()

freq={}
for name in names:
    freq[name]=freq.get(name,0)+1

print(freq)
```

Output:

```
{'Shekhar': 1, 'Sagar': 1, 'Dipak': 1, 'Priyanka': 1, 'Roshni': 1, 'Suraj': 1, 'Akash': 1}
```

8. **String Analysis o For each name, print:**
   - **Length of the name**
   - **First and last character**
   - **Whether it is a palindrome**

```
with open("student.txt") as f:
    names=f.read().splitlines()

for name in names:
    print(name,len(name),name[0],name[-1],name.lower()==name.upper()[::-1])
```

Output:

```
Shekhar 7 S r False
Sagar 5 S r False
Dipak 5 D k False
Priyanka 8 P a False
Roshni 6 R i False
Suraj 5 S j False
Akash 5 A h False
```

9. **File + List + String Combined Challenge**

- **Read names from file → list.**
- **Remove duplicates.**
- **Sort by length (shortest to longest).**
- **Write results into processed_names.txt**

```python
with open("student.txt") as f:
    names=f.read().splitlines()

names=sorted(names,key=len)

with open("processed.txt","w") as f:
    f.write("\n".join(names))
```

Output:

processed.txt

```
1   Sagar
2   Dipak
3   Suraj
4   Akash
5   Roshni
6   Shekhar
7   Priyanka
```

10.

- **Read names from file.**
- **Group them by starting letter (e.g., all names starting with A together).**
- **Save each group into separate files (A_names.txt, B_names.txt, etc.).**

```python
with open("student.txt") as f:
    names=f.read().splitlines()

groups={}
for name in names:
    letter=name[0].upper()
    groups.setdefault(letter,[]).append(name)

for letter,group in groups.items():
    with open(f"{letter}.txt","w") as f:
        f.write("\n".join(group))
```

Output:

A.txt

```
1    Akash
```

D.txt

```
1    Dipak
```

P.txt

```
1    Priyanka
```

R.txt

```
1    Roshni
```

S.txt

```
1    Shekhar
2    Sagar
3    Suraj
```